Proofs with Sets

Membership, subsets and set equality

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Source

Adapted from Kameron Decker Harris's Fall 2026 lecture notes "Proofs Involving Sets", which are typed transcriptions of Scott Wehrwein's Spring 2025 handwritten lecture notes. These come from those notes:

- what to show to prove that an element is in a set given in set-builder notation;
- the direct and contrapositive outlines of a subset proof;
- proving set equality by proving that each set is a subset of the other;
- the theorem that a nonempty factor cancels from both sides of an equation between Cartesian products, with the proof of its first half.

The exercises, the second half of that proof, the membership test for a set built with an expression, and the section on disproving a claim about sets are new.

Proving membership

Exercise 1 (In or out)**.** Let $T = \{3k + 2 : k \in \mathbb{Z}\}$ (Definition: Set-builder notation with an expression). Prove that $17 \in T$ and that $19 \notin T$.

Solution

Solution 1. $17 \in T$**.** An element of T is a number of the form $3k + 2$ with k an integer. Take $k = 5$, which is an integer. Then $3k + 2 = 3 \cdot 5 + 2 = 17$, so $17 \in T$.
$19 \notin T$**.** Suppose, for contradiction, that $19 \in T$ (Definition: Proof by contradiction). Then $19 = 3k + 2$ for some integer k. Subtracting 2 from both sides gives $3k = 17$, so $k = 17/3$. But $17/3$ is not an integer, since $3 \cdot 5 = 15 < 17 < 18 = 3 \cdot 6$. This contradicts

that k is an integer, so $19 \notin T$.

Remark 1 (What to show for membership). To prove that a is an element of a set, show that a passes the set's test (Definition: Set-builder notation, Definition: Set-builder notation with an expression):

- for $S = \{x : P(x)\}$, the same set-builder notation with no set A named, show that $P(a)$ is true;
- for $S = \{x \in A : P(x)\}$, show that $a \in A$ and that $P(a)$ is true;
- for $S = \{f(k) : k \in A\}$, find some $k \in A$ with $f(k) = a$.

The last case is an existence proof, so it usually names the k (Definition: Constructive proof). To prove that $a \notin S$, prove the negation of the test, often by contradiction, as in Solution 1.

Example 1 (A root of a polynomial)**.** Let $S = \{x \in \mathbb{R} : 2x^2 - x = 0\}$. Then $\frac{1}{2} \in S$.
The set has the form $\{x \in A : P(x)\}$, so there are two things to show (Remark 1). First, $\frac{1}{2}$ is a real number. Second,

$$\begin{aligned} 2\left(\frac{1}{2}\right)^2 - \frac{1}{2} &= 2 \cdot \frac{1}{4} - \frac{1}{2} \\ &= \frac{1}{2} - \frac{1}{2} \\ &= 0. \end{aligned}$$

Proving a subset

Exercise 2 (Where a quadratic is negative)**.** Prove that $\{x \in \mathbb{R} : x^2 - 3x + 2 < 0\} \subseteq \{x \in \mathbb{R} : 0 < x < 3\}$ (Definition: Subset).
Hint: factor $x^2 - 3x + 2$.

Solution

Solution 2. Let $x \in \{x \in \mathbb{R} : x^2 - 3x + 2 < 0\}$. Then x is a real number, and $x^2 - 3x + 2 = (x-1)(x-2) < 0$. We show that $1 < x < 2$, by ruling out the other two cases (Definition: Proof by cases).

- If $x \leq 1$, then $x - 1 \leq 0$ and $x - 2 < 0$, so $(x-1)(x-2) \geq 0$, since the product of two numbers that are not positive is not negative. That contradicts $(x-1)(x-2) < 0$.

- If $x \geq 2$, then $x - 1 > 0$ and $x - 2 \geq 0$, so $(x-1)(x-2) \geq 0$, again a contradiction.

So $1 < x < 2$, and so $0 < x < 3$. So $x \in \{x \in \mathbb{R} : 0 < x < 3\}$.

Remark 2 (The shape of a subset proof). By Definition: Subset, $A \subseteq B$ means the conditional "if $a \in A$, then $a \in B$" is true for every a. So a proof of $A \subseteq B$ is a proof of that conditional, either directly (Definition: Direct proof):

> Suppose $a \in A$.
> ⋮
> Therefore $a \in B$.
> So $A \subseteq B$.

or by contrapositive (Definition: Proof by contrapositive):

> Suppose $a \notin B$.
> ⋮
> Therefore $a \notin A$.
> So $A \subseteq B$.

The element a is arbitrary: the proof may use only that $a \in A$ (or that $a \notin B$), nothing else about it.

Example 2 (A subset proof by contrapositive)**.** Claim: $\{x \in \mathbb{R} : x^3 + x > 0\} \subseteq \{x \in \mathbb{R} : x > 0\}$.
We use the contrapositive outline of Remark 2. Suppose $a \notin \{x \in \mathbb{R} : x > 0\}$. If a is not a real number, then a is not in the left-hand set either, since every element of that set is a real number. Otherwise a is a real number with $a \leq 0$. Then $a^3 = a \cdot a^2 \leq 0$, since $a \leq 0$ and $a^2 \geq 0$, so

$$a^3 + a \leq 0 + 0 = 0.$$

So $a \notin \{x \in \mathbb{R} : x^3 + x > 0\}$.
The direct outline would start from "$a^3 + a > 0$", and would have to get from a cubic back to a, which is harder.

Theorem 1 (Subsets of subsets)**.** *Let A, B and C be sets. If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.*

Proof

Proof. Suppose $A \subseteq B$ and $B \subseteq C$, and let $a \in A$. Since $A \subseteq B$, $a \in B$ (Definition: Subset). Since $B \subseteq C$, $a \in C$. So every element of A is in C, and $A \subseteq C$. □

Example 3 (A chain of subsets)**.** Solution 2 shows that $\{x \in \mathbb{R} : x^2 - 3x + 2 < 0\} \subseteq \{x \in \mathbb{R} : 0 < x < 3\}$, and every real number between 0 and 3 is positive, so $\{x \in \mathbb{R} : 0 < x < 3\} \subseteq \{x \in \mathbb{R} : x > 0\}$. By Theorem 1, $\{x \in \mathbb{R} : x^2 - 3x + 2 < 0\} \subseteq \{x \in \mathbb{R} : x > 0\}$.

Proving set equality

Exercise 3 (Absorbing an intersection)**.** Let A and B be sets. Prove that $A \cup (A \cap B) = A$ (Definition: Union, Definition: Intersection).
Hint: prove that each side is a subset of the other.

Solution

Solution 3. $A \cup (A \cap B) \subseteq A$**.** Let $x \in A \cup (A \cap B)$. Then $x \in A$ or $x \in A \cap B$. In the first case, $x \in A$. In the second case, $x \in A$ and $x \in B$, so again $x \in A$. Either way, $x \in A$ (Definition: Proof by cases).
$A \subseteq A \cup (A \cap B)$**.** Let $x \in A$. Then "$x \in A$ or $x \in A \cap B$" is true, since its first part is true (Definition: Or), so $x \in A \cup (A \cap B)$.
Each side is a subset of the other, so an element is in one side exactly when it is in the other, and the two sets are equal (Definition: Set).

Theorem 2 (Equal sets are subsets of each other)**.** *Let A and B be sets. Then $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$.*

Proof

Proof. By Definition: Set, $A = B$ means that every x has $x \in A \Leftrightarrow x \in B$. By Theorem: A biconditional is two conditionals, that biconditional is the same as "$x \in A \Rightarrow x \in B$, and $x \in B \Rightarrow x \in A$". A conjunction is true for every x exactly when each of its two parts is true for every x. The first conditional, for every x, is $A \subseteq B$ (Definition: Subset), and the second, for every x, is $B \subseteq A$. □

Remark 3 (The shape of a set equality proof). By Theorem 2, a proof of $A = B$ has two parts, each a subset proof (Remark 2):

($\subseteq$) Suppose $a \in A$.
⋮
Therefore $a \in B$.

($\supseteq$) Suppose $a \in B$.
⋮
Therefore $a \in A$.

So $A = B$.

This is the "if and only if" outline of Remark: The shape of an "if and only if" proof, applied to $a \in A$ and $a \in B$.

Example 4 (Two kinds of odd number)**.** Claim: $\{4k+1 : k \in \mathbb{Z}\} \cup \{4k+3 : k \in \mathbb{Z}\} = \{2m+1 : m \in \mathbb{Z}\}$.
($\subseteq$) Let x be in the left-hand side. If $x = 4k+1$ for some integer k, then $x = 2(2k)+1$. If $x = 4k+3$ for some integer k, then $x = 2(2k+1)+1$. Both $2k$ and $2k+1$ are integers (Remark: Facts about integers used without proof), so either way x is in the right-hand side.
($\supseteq$) Let $x = 2m+1$ for some integer m. The integer m is even or odd (Remark: Facts about integers used without proof).

- If m is even, then $m = 2j$ for some integer j (Definition: Even integer), so $x = 2(2j)+1 = 4j+1$, and $x \in \{4k+1 : k \in \mathbb{Z}\}$.
- If m is odd, then $m = 2j+1$ for some integer j (Definition: Odd integer), so $x = 2(2j+1)+1 = 4j+3$, and $x \in \{4k+3 : k \in \mathbb{Z}\}$.

Either way, x is in the left-hand side. So the two sets are equal (Theorem 2).

Theorem 3 (A subset is its own intersection)**.** *Let A and B be sets. Then $A \subseteq B$ if and only if $A \cap B = A$.*

ℹ Proof

Proof. ($\Rightarrow$) Suppose $A \subseteq B$. We show $A \cap B = A$ with Theorem 2. If $x \in A \cap B$, then $x \in A$ (Definition: Intersection). If $x \in A$, then $x \in B$, since $A \subseteq B$, so $x \in A \cap B$.
($\Leftarrow$) Suppose $A \cap B = A$, and let $x \in A$. Then $x \in A \cap B$, since the two sets are equal, so

$x \in B$. So $A \subseteq B$. □

Example 5 (Intersecting with a larger set)**.** Let $B = \{1, 2, 3, 4\}$.

- $\{2, 4\} \subseteq B$, and $\{2, 4\} \cap B = \{2, 4\}$, as Theorem 3 says.
- $\{2, 5\} \not\subseteq B$, since $5 \notin B$, and $\{2, 5\} \cap B = \{2\}$, which is not $\{2, 5\}$.

Theorem 4 (Cancelling a factor of a Cartesian product)**.** *Let A, B and C be sets with $C \neq \emptyset$. If $A \times C = B \times C$, then $A = B$ (Definition: Cartesian product).*

Proof

Proof. Suppose $A \times C = B \times C$. Since $C \neq \emptyset$, there is some $c \in C$ (Definition: Empty set).
($\subseteq$) Let $a \in A$. Then $(a, c) \in A \times C$. Since $A \times C = B \times C$, also $(a, c) \in B \times C$, so $(a, c) = (b, c')$ for some $b \in B$ and $c' \in C$ (Definition: Cartesian product). Two ordered pairs are equal only when their first entries are equal (Definition: Ordered pair), so $a = b$, and $a \in B$.
($\supseteq$) Let $b \in B$. Then $(b, c) \in B \times C = A \times C$, so $b \in A$, by the same steps with A and B swapped.
So $A = B$ (Theorem 2). □

Example 6 (Why C must not be empty)**.** The condition $C \neq \emptyset$ in Theorem 4 is needed. Let $A = \{1\}$, $B = \{2\}$ and $C = \emptyset$. A pair in $A \times C$ would need a second entry in C, and C has none, so $A \times C = \emptyset$, and likewise $B \times C = \emptyset$. So $A \times C = B \times C$, but $A \neq B$. The proof uses the condition in its first step, to find an element $c \in C$.

Disproving a claim about sets

Exercise 4 (Regrouping differences)**.** Show that the claim "for all sets A, B and C, $A - (B - C) = (A - B) - C$" is false (Definition: Set difference).

Solution

Solution 4. A counterexample (Definition: Counterexample) is one choice of A, B and C for which the two sides differ. Let $A = B = C = \{1\}$. Then

$$\begin{aligned} A - (B - C) &= \{1\} - (\{1\} - \{1\}) \\ &= \{1\} - \emptyset \\ &= \{1\}, \end{aligned}$$

while

$$\begin{aligned} (A - B) - C &= (\{1\} - \{1\}) - \{1\} \\ &= \emptyset - \{1\} \\ &= \emptyset. \end{aligned}$$

The two sides differ, so the claim is false.

Remark 4 (Disproving a claim about all sets). A claim of the form "for all sets A, B, …, some equation holds" is disproved by one choice of sets for which it fails (Definition: Disproof). Small sets, such as $\emptyset$ and $\{1\}$, are a good place to look. A Venn diagram (Definition: Venn diagram) can suggest where to look: shade each side, find a region shaded on one side only, and put an element in that region.

Example 7 (A union does not cancel)**.** Claim to disprove: for all sets A, B and C, if $A \cup B = A \cup C$, then $B = C$.
Let $A = \{1\}$, $B = \{1\}$ and $C = \emptyset$. Then $A \cup B = \{1\}$ and $A \cup C = \{1\}$, so $A \cup B = A \cup C$, but $B \neq C$. So the claim is false.